10.11 Hydrogenated fats

Hydrogenation (or, more accurately, "partial hydrogenation," as the process is incomplete) is the forced chemical addition of hydrogen into omega-6 polyunsaturated oils to make them hard at room temperatures, primarily as a cheaper and less perishable substitute for butter. The liquid fat becomes a solid fat and the unsaturated fatty acid contents decrease as a result of hydrogenation. Common hydrogenated fats include hydrogenated or partially hydrogenated cottonseed, palm, soy and corn oils, but theoretically almost any polyunsaturated oil can be hydrogenated. During the process of hydrogenation, hydrogen is added to the unsaturated linkage with nickel as catalyst.

A major health concern during the hydrogenation process is the production of trans fats. Trans fats are the result of a side reaction with the catalyst of the hydrogenation process. This is the result of an unsaturated fat which is normally found as a cis isomer converts to a trans isomer of the unsaturated fat. Isomers are molecules that have the same molecular formula but are bonded together differently. A cis isomer has the hydrogens on the same side, whereas a trans isomer has hydrogen atoms on the opposite side. Due to the added energy from the hydrogenation process, the activation energy is reached to convert the cis isomers of the unsaturated fat to a trans isomer of the unsaturated fat .

Although trans fats are edible, consumption of trans fats has been shown to increase the risk of coronary artery disease in part by raising levels of the lipoprotein LDL (often referred to as "bad cholesterol"), lowering levels of the lipoprotein HDL (often referred to as "good cholesterol"), increasing triglycerides in the bloodstream and promoting systemic inflammation. Trans fat are found in margarine, vanaspathi, baked goods such as doughnuts, pastries, cookies, deep fried foods like fried chicken and French-fried potatoes, microwave popcorn, snack chips, processed foods and confectionery fats.

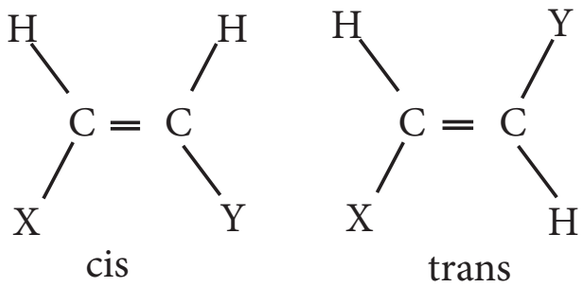


Fig 10.19: Cis and trans fats

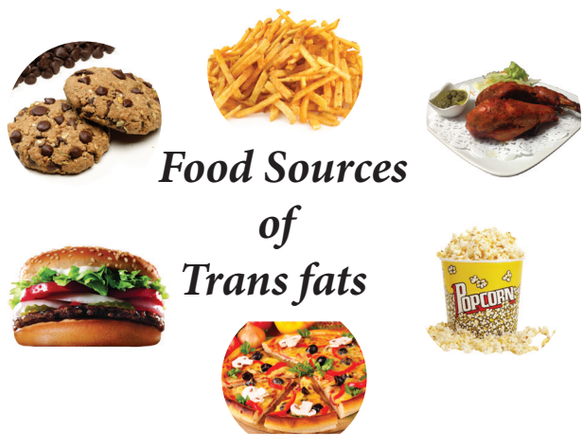


Fig 10.20: Food sources of trans fats

ACTIVITY - 5

1) Display food sources of GOOD and BAD fats.
2) What on the foods containing trans fats which you frequently consume and list their harmful effects.

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